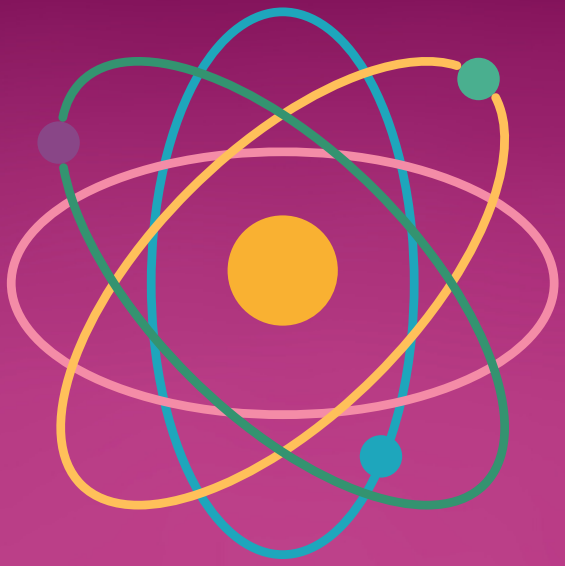
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

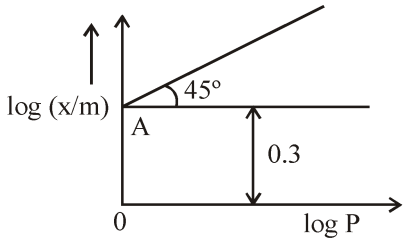
EXERCISE

Surface Chemistry

English Medium

EXERCISE # S-I

1. Graph between $\log\left(\frac{x}{m}\right)$ vs log P is provided for adsorption of NH_3 gas on metal surface. Calculate weight of NH_3 gas adsorbed by 50 gm of metal surface at 2 atm pressure



SC0001

2. When a graph is plotted between *log x/m* and *log P* (where P is in atm), it is straight line with an angle of 45º and intercept 0.3010 on y-axis, what will be the amount of gas adsorbed per gm of adsorbent at pressure 0.3 atm (log2 = 0.3010) **SC0002**

3. In vessel O_2 gas undergoes physisorption on adsorbent surface causing its partial pressure to decrease from 6 atm to 'P' atm in a 8.21 L vessel at 500K. If total number of sites is 2×10^{23} on adsorbent surface and every site accommodates effectively three O_2 molecules find P(atm).

$[N_A = 6 \times 10^{23}]$ **SC0003**

4. 1.25 mg of Gum Arabica (Gold Number = 0.25) is added to 50 ml of standard gold sol. What maximum volume (in ml) of 10% NaCl solution can be added to this gold sol without causing coagulation. **SC0004**

5. Gold number of haemoglobin is 0.3. Calculate weight (in mg) of haemoglobin for 100 ml of gold sol so that gold sol is not coagulated by 10 ml of 10% NaCl solution. **SC0005**

6. A soap ($C_{17}H_{35}$ COONa) solution becomes a colloidal sol at a concentration of 1.2×10^{-3} M. On the average, 2.4×10^{13} colloidal particles are present in 1 mm^3. What is the average number of stearate ions in one colloidal particle (micelle) : **Take : Avogadro's number = 6×10^{23}** . **SC0006**

7. How many of the following phenomenon are observed due to process of coagulation.

(A) Delta formation at the meeting point of river and sea.

(B) Blue colour of sky.

(C) Precipitation of coal ash from smoke in chimeney's.

(D) destruction of lyophobic solution.

(E) Brownian motion.

(F) Cleaning action of detergents.

(G) Artificial rain.

(H) Use of alums in cleaning water. **SC0007**

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8. Calculate the number of sols which are negatively charged.

$Fe_2O_3.xH_2O$ sol	Sb_2S_3 sol
Ag sol	Cu sol
Blood	basic dye
sol of clay	sol of starch
sol of sodium stearate	sol of sodium laurylsulphate
sol of charcoal.	

SC0008

9. In order to cause coagulation of 200 ml of gold sol, 585 ml of 1% w/w NaCl solution having density 1.2 gm/ml was required. What will be the coagulation value of NaCl?

[express answer in milli-moles /litre] **SC0009**

10. Molecular formula of starch can be represented as $(C_6H_{10}O_5)_n$. If the gold number of one such starch molecule is 6.48 and 0.01 millimoles of the above starch are required to be added to 10ml of gold sol to prevent coagulation by 1 ml of 10% sodium chloride solution, then calculate the value of 'n'.

SC0010

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EXERCISE # S-II

1. In a vessel O_2 gas molecules were adsorbed on solid surface causing its partial pressure to decreases from 1 atm to 0.5 atm.

Given : $N_A = 6 \times 10^{23}$

Volume of vessel = 2.24 L

Temperature = 273 K

Total area of solid surface = 10^2 cm^2

Number of active sites per unit area = 10^{24} m^{-2}

Find number of O_2 molecules adsorbed per active site. **SC0011**

2. A 0.03 M of an acid solution in benzene is dropped on water surface, the benzene evaporates and the acid forms a monomolecular film of solid type. What volume (in ml) of the above solution would be required to cover a 54000cm^2 surface area of water with monomolecular layer of acid. Area covered by single acid molecule is 0.3 nm^2 ($N_A = 6 \times 10^{23}$) **SC0012**

3. The diameter of a colloidal particle is 5000 Å. If the density of substance formed , dispersed phase, is 4gm/cc, find the value of surface area per unit mass of colloidal particle in (m^2/gm).

SC0013

4. A container contains 1 litre, 2M solution of benzene in ether. A piece of 3 kg charcoal is dipped in the solution. Molecules of benzene get adsorbed on the surface of charcoal and form monolayer. The molarity of resulting solution decreases to 1M. If surface area available for adsorption on charcoal $12\sqrt{3}$ cm^2/gm. Then find distance (in pm) between two adjacent carbon atoms in a benzene molecule.

(Assume: Shape of benzene molecule perfectly hexagonal.)

[Use: $N_A = 6 \times 10^{23}$] **SC0014**

5. 112cm^3 hydrogen gas is adsorbed uniformly at the surface of 5 gm Palladium at 273°C and 2 atm. If the effective surface area of each hydrogen molecule is 0.4 nm^2, then the specific surface area of Palladium is : **SC0015**

6. The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time upto which a N_2 molecule may remain adsorbed at Pt-surface at 400K is

[Given: Pre-exponential factor, $A = 1.25 \times 10^8\ s^{-1}$;

Activation energy of desorption = 16Kcal, $e^{20} = 5 \times 10^8$] **SC0016**

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7. A colloid prepared by the addition of KI to $AgNO_3$ solution is purified using dialysis. Find the minimum mass in grams of an electrolyte 'AB' (GMM = 60) required to completely coagulate 1ℓ of the aforementioned colloid

Given :

Active ion causing flocculation	Flocculation value(mmole/l)
A^+	50
B^-	100

SC0017

8. At 70K, the adsorption of N_2 gas at iron surface obeys Freundlich adsorption isotherm. The experimental data collected is

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

Where $\frac{x}{m}$ is the mass (in gm) of N_2 gas adsorbed per gm of iron at P bar pressure. The moles of N_2 gas adsorbed per gm of iron at 36 bar and 70 K, is **SC0018**

9. 1.9×10^{-4} gm of the metal having density 19 gm/ml is dispersed in one litre of water to give a sol having spherical metal particles of radius 10 nm. The approximate number of metal sol particles per cm^3 of the sol is given by : **SC0019**

10. If unit mass of a solid, taken as cube of volume 8 cm^3, is powdered into identical 10^{12} cubes, then the specific surface area of the solid increased by : **SC0020**

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EXERCISE # O-I

Select the correct alternative. (Only one is correct)

1. Which gas will be adsorbed on a solid to greater extent.
(A) A gas having non polar molecule
(B) A gas having highest critical temperature (T_c)
(C) A gas having lowest critical temperature.
(D) A gas having highest critical pressure. **SC0021**

2. The heat of physisorption lie in the range of
(A) 1 – 10 kJ mol^{-1} (B) 20 to 40 kJ mol^{-1}
(C) 40 to 200 kJ mol^{-1} (D) 200 to 400 kJ mol^{-1} **SC0022**

3. Adsorption is multilayer in case of
(A) physical adsorption (B) chemisorption
(C) in both (D) none of the these **SC0023**

4. Reversible adsorption is
(A) chemical adsorption (B) physical adsorption
(C) both (D) none **SC0024**

5. An emulsion is a colloidal system of
(A) two solids (B) two liquids
(C) one gas and one solid (D) one gas and one liquid **SC0025**

6. The nature of bonding forces in chemisorption
(A) purely physical such as Van Der Waal's forces
(B) purely chemical
(C) both chemical and physical simultaneously.
(D) none of these **SC0026**

7. The Tyndall effect associated with colloidal particles is due to
(A) presence of electrical charges (B) scattering of light
(C) absorption of light (D) reflection of light **SC0027**

8. Which one of the following is not applicable to chemisorption?
(A) Its heat of adsorption is high (B) It takes place at high temperature
(C) It is reversible (D) It forms mono-molecular layers **SC0028**

9. In the colloidal state the particle size ranges
(A) below 1 nm (B) between 1 nm to 1000 nm
(C) more than 1000 nm (D) none of the above **SC0029**

10. Colloids can be purified by
(A) condensation (B) peptization (C) coagulation (D) dialysis **SC0030**

11. Milk is an example of
(A) emulsion (B) suspension (C) foam (D) sol. **SC0031**

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12. Colloidal particles in a sol. can be coagulated by
(A) heating (B) adding an electrolyte
(C) adding oppositely charged sol (D) any of the above methods **SC0022**

13. Emulsifier is an agent which
(A) accelerates the dispersion (B) homogenizes an emulsion
(C) stabilizes an emulsion (D) aids the flocculation of an emulsion **SC0033**

14. Fog is a colloidal system of
(A) gas in liquid (B) liquid in gas (C) gas in gas (D) gas in solid **SC0034**

15. Given below are a few electrolytes, indicate which one among them will bring about the coagulation of a gold sol. quickest and in the least of molar concentration?
(A) NaCl (B) $MgSO_4$ (C) $Al_2(SO_4)_3$ (D) $K_4[Fe(CN)_6]$ **SC0035**

16. When a lyophobic colloidal solution is observed , we can see
(A) light scattered by colloidal particle
(B) size of the colloidal particle
(C) shape of the colloidal particle
(D) relative size of the colloidal particle **SC0036**

17. The electrical charge on a colloidal particle is indicated by
(A) Brownian movement (B) electrophoresis
(C) ultra microscope (D) molecular sieves **SC0037**

18. The minimum concentration of an electrolyte required to cause coagulation of a sol is called
(A) flocculation value (B) gold number (C) protective value (D) none of these **SC0038**

19. Smoke precipitator works on the principle of
(A) distribution law (B) neutralization of charge on colloids
(C) Le-Chaterlier's principle (D) addition of electrolytes **SC0039**

20. Which one of following statements is not correct in respect of lyophilic sols?
(A) There is a considerable interaction between the dispersed phase and dispersion medium
(B) These are quite stable and are not easily coagulated
(C) They need stabilizing agent
(D) The particle are hydrated **SC0040**

21. As_2S_3 sol is
(A) positive colloid (B) negative colloid (C) neutral colloid (D) none of the above **SC0041**

22. At the critical micelle concentration (CMC) the surfactant molecules
(A) decompose (B) dissociate
(C) associate (D) become completely soluble **SC0042**

23. Small liquid droplets dispersed in another liquid is called
(A) suspension (B) emulsion (C) gel (D) true solution **SC0043**

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24. The process which is catalysed by one of the product is called
(A) acid-base catalysis (B) autocatalysis
(C) negative catalysis (D) homogeneous catalysis **SC0044**

25. Tyndal effect would be observed in a
(A) solution (B) solvent (C) precipitate (D) colloidal sol. **SC0045**

26. A liquid is found to scatter a beam of light but leaves no residue when passed through the filter paper. The liquid can be described as
(A) a suspension (B) oil (C) a colloidal sol. (D) a true solution **SC0046**

27. The ability of an ion to bring about coagulation of a given colloid depends upon
(A) its charge (B) the sign of the charge alone
(C) the magnitude of the charge (D) both magnitude and sign of charge **SC0047**

28. An arsenious sulphide sol. carries a negative charge. The maximum precipitating power of this sol. is possessed by
(A) K_2SO_4 (B) $CaCl_2$ (C) Na_3PO_4 (D) $AlCl_3$ **SC0048**

29. Which of the following is an example of associated colloid?
(A) Protein + water (B) Soap + water (C) Rubber + benzene (D) $As_2O_3 + Fe(OH)_3$ **SC0049**

30. Although nitrogen does not adsorb on surface at room temperature, it adsorbs on surface at 83K. Which one of the following statements is correct -
(A) At 83K, there is formation of monomolecular layer
(B) At 83K, there is formation of multimolecular layer
(C) At 83K, nitrogen molecules are held by chemical bonds
(D) At 83K, nitrogen is adsobed as atoms. **SC0050**

31. Gold number of a lyophilic sol is such a property that :
(A) The larger its value, the greater is the peptizing power
(B) The lower its value, the greater is the peptizing power
(C) The lower its value, the greater is the protecting power
(D) The larger its value, the greater is the protecting power **SC0051**

32. Which of the following statements is incorrect regarding physisorptions ?
(A) Under high pressure it results into multi molecular layer on adsorbent surface
(B) Enthalpy of adsorption ($\Delta H_{adsorption}$) is low and positive
(C) It occurs because of Van der Waal's forces
(D) More easily liquefiable gases are adsorbed readily **SC0052**

33. The volume of a colloidal particle V_C, volume of a solute particle in a true solution V_t, the volume of suspension particle is V_S can be arranged
(A) $V_C = V_t = V_S$ (B) $V_S < V_C < V_t$
(C) $V_S > V_C > V_t$ (D) $V_C > V_S > V_t$ **SC0053**

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34. Which of the following is not the property of physi-sorption -

(A) Highly specific in nature (B) Reversible

(C) Multilayer (D) Exothermic **SC0054**

35. Arrange the following electrolytes in the increasing order of coagulation power for the coagulation of As_2S_3 sol -

(I) Na_3PO_4 (II) $MgCl_2$ (III) $AlCl_3$

(A) I > II > III (B) I < II < III (C) I = III < II (D) III < I < II

SC0055

36. The gas, which is most readily adsobred on the surface of activated charcoal is -

(A) N_2 (B) H_2 (C) O_2 (D) SO_2 **SC0056**

37. The migration of colloidal particles under the influence of an electrical field is known as

(A) electro osmosis (B) electrophoresis

(C) electrodialysis (D) None **SC0057**

38. Which reaction show the use of heterogenous catalyst

(A) $2SO_2\,(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

(B) $2SO_2\,(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$

(C) $CH_3COOCH_3(l) + H_2O(l) \xrightarrow{HCl(l)} CH_3COOH(aq.) + CH_3OH(aq.)$

(D) All of the above **SC0058**

39. Promoters and Poison are the substance use in chemical reaction which

(A) Enhance and decrease the activity of catalyst respectively

(B) Decrease and enhance the activity of catalyst respectively

(C) Do not have any effect on catalyst

(D) Can be used in place of catalyst whenever required **SC0059**

40. Peptization process may be defined as

(A) Formation of precipitate by adding two ionic solution

(B) Conversion of colloidal sol into precipitate

(C) Conversion of precipitate into colloidal sol

(D) Enrichment precipitate from solution **SC0060**

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EXERCISE # O-II

(More than one may be correct)

1. Which of the following is/are correct statements

(A) Hardy Schulz rule is related to coagulation

(B) Brownian movement and Tyndall effect are shown by colloids

(C) When liquid is dispersed in liquid, it is called gel.

(D) Gold number is a measure of protective power of lyophillic colloid. **SC0061**

2. Which statements is/are correct?

(A) Physical adsorption is multilayer non-directional and non specific

(B) Chemical adsorption is generally monolayer and specific in nature

(C) Physical adsorption is due to free valence of atoms

(D) Chemical adsorption is stronger than physical adsorption **SC0062**

3. Which statement/s is/are correct

(A) A solution is prepared by addition of excess of $AgNO_3$ solution in KI solution. The charge likely to develop on colloidal particle is positive.

(B) The effects of pressure on physical adsorption is high if temperature is low.

(C) Gold number is the index for extent of gold plating done.

(D) None **SC0063**

4. Colloidal solution can be purified by

(A) Dialysis (B) Electrodialysis (C) Electrophoresis (D) Ultrafiltration **SC0064**

5. Which of the following is not lyophillic

(A) Gelatin sol (B) Silver sol (C) Sulphur sol (D) As_2S_3 sol **SC0065**

6. Colloidal Gold can be prepared by

(A) Bredig's are method (B) Reduction of $AuCl_3$

(C) Hydrolysis (D) Peptization **SC0066**

7. On adding $AgNO_3$ solution into KI solution , colloidal sol can be obtained from.

(A) 100 mL of 0.1 M $AgNO_3$ + 100 of 0.1 M KI

(B) 100 mL of 0.1 M $AgNO_3$ + 100 of 0.2 M KI

(C) 100 mL of 0.2 M $AgNO_3$ + 100 of 0.1 M KI

(D) 100 mL of 0.15 M $AgNO_3$ + 100 of 0.25 M KI **SC0067**

Question No. 8 to 10 (3 questions)

Whenever a mixture of gases is allowed to come in contact with a particular adsorbent under the same conditions, the more strong adsorbate is adsorbed to greater extent irrespective of its amount present, e.g. H_2O is adsorbed to more extent on silica gel than N_2 and O_2. This shows that some adsorbates are preferentially adsorbed. It is also observed that preferentially adsorbable adsorbate can displace a weakly adsorbed substance from the surface of an adsorbent.

8. Which of the following gases is adsorbed to maximum extent:

(A) He (B) Ne (C) Ar (D) Xe **SC0068**

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9. Which of the gas can displace remaining all the gases

(A) O_2 (B) N_2 (C) CO (D) H_2 **SC0069**

10. When temperature is increased

(A) extent of adsorption increases

(B) extent of adsorption decreases

(C) no effect on adsorption

(D) extent of adsorption first decreases, then increases **SC0070**

Question No. 11 to 12(2 questions)

In macromolecular type of colloids, the dispersed particles are themselves large molecules (usually polymers). Since these molecules have dimensions comparable to those of colloidal particles, their dispersions are called macromolecular colloids. Most lyophilic sols belong to this category. There are certain colloids which behave as normal strong electrolytes at low concentrations, but exhibit colloidal properties at higher concentrations due to the formation of aggregated particles. These are known as micelles or associated colloids. Surface active agents like soaps and synthetic detergents belong to this class.

CMC increases with the total surfactant concentration. At concentration higher than CMC, they form extended parallel sheets known as **lamellar micelles** which resemble biological membranes. With two molecules thick, the individual molecule is perpendicular to the sheets such that hydrophilic groups are on the outside in aqueous solution and on the inside is a non-polar medium.

In concentrated solutions, micelles take the form of long cylinders packed in hexagonal arrays and are called lytotropic mesomorphs.

In an aqueous solution (polar medium), the polar group points towards the periphery and the hydrophobic hydrocarbon chains point towards the centre forming the core of the micelle.

-They are capable of forming ions

-Molecules of soaps and detergents consist of lyophilic as well as lyophilic parts which associate together to form micelles.

-Micelles may contain as many as 100 molecules or more.

11. Select incorrect statement(s):

(A) Surface active agents like soaps and synthetic detergents always form micelles in water

(B) Soaps are emulsifying agents

(C) $C_{17}H_{35}$ (hydrocarbon part) and $-COO^-$ (carboxylate) part of stearate ion ($C_{17}H_{35}COO^-$) both are hydrophobic

(D) All are incorrect statements **SC0071**

12. Cleansing action of soap occurs because:

(A) oil and grease can be absorbed into the hydrophobic centres of soap micelles and washed away

(B) oil and grease can be absorbed into hydrophilic centres of soap micelles and washed away

(C) oil and grease can be absorbed into both hydrophilic and hydrophobic centres but not washed away

(D) cleansing action is not related to micelles **SC0072**

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Question No.13 to 15 (3 questions)

The protective power of the lyophilic colloids is expressed in terms of gold number a term introduced by Zsigmondy.Gold number is the number of milligram of the protective colloid which prevent the coagulation of 10 ml of red gold sol. when 1 ml of a 10 percent solution of sodium chloride is added to it. Thus, smaller the gold number of lyophilic colloid, the greater is the protective power.

13. On addition of one mL solution of 10% NaCl to 10 mL of red gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is

(A) 0.025 (B) 0.25 (C) 2.5 (D) 25 **SC0073**

14. Which of the following statement(s) is/are correct

(A) Higher the gold number, more protective power of colloid

(B) Lower the gold number, more the protective power

(C) Higher the coagulation value, more the coagulation power

(D) Lower the coagulation value, higher the coagulation power **SC0074**

15. Gold number gives an indication of

(A) protective nature of colloids

(B) purity of gold in suspension

(C) the charge on a colloidal solution of gold

(D) g-mole of gold per litre **SC0075**

Question No. 16 & 19 (4 questions)

These questions consist of two statements each, printed as assertion and reason, while answering these questions you are required to choose any one of the following responses.

(A) If both assertion and reason are true and the reason is a correct explanation of assertion.

(B) If both assertion and reason are true but reason is not a correct explanation of assertion.

(C) If assertion is true but the reason is false.

(D) If assertion is false but the reason is true.

16. **Assertion :** When $AgNO_3$ is treated with excess of potassium iodide, colloidal particles gets attracted towards anode.

Reason : Precipitate adsorb common ions (excess) and thus become charged. **SC0076**

17. **Assertion :** For adsorption ΔG, ΔH, ΔS all have negative values

Reason : Adsorption is a spontaneous exothermic process in which randomness decreases due to force of attraction between adsorbent and adsorbate. **SC0077**

18. **Assertion :** A gas with higher critical temperature gets adsorbed to more extent than a gas with lower critical temperature. **SC0078**

Reason : The easily liquifiable gases get adsorbed to less extent.

19. **Assertion :** Micelles are formed by surfactant molecules at concentrations above the critical micellar concentration (CMC) and temperature above Kraft temperature.

Reason : The molar conductivity of a solution having surfactant molecules decreases sharply at the CMC. **SC0079**

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LIST TYPE :

20.

	List-I		**List-II**
(P)	Inversion of cane sugar	(1)	Diastase
(Q)	Conversion of starch into maltose	(2)	Lactor bacilli enzyme
(R)	Conversion of glucose into ethyl alcohol	(3)	Invertase
(S)	Conversion of milk into curd	(4)	Zymase

Code :

	P	**Q**	**R**	**S**
(A)	3	4	1	2
(B)	3	4	2	1
(C)	3	1	4	2
(D)	2	4	1	2

SC0080

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EXERCISE # J-MAIN

1. Adsorption of a gas on a surface follows Freundlich adsorption isotherm. Plot of $\log\frac{x}{m}$ versus log p gives a straight line with slope equal to 0.5, then: **[JEE-MAIN-2017]**

($\frac{x}{m}$ is the mass of the gas adsorbed per gram of adsorbent)

(1) Adsorption is proportional to the square of pressure.

(2) Adsorption is independent of pressure.

(3) Adsorption is proportional to the pressure.

(4) Adsorption is proportional to the square root of pressure. **SC0088**

2. Which one of the following is not a property of physical adsorption **[JEE-MAIN-2018]**

(1) Unilayer adsorption occurs

(2) Greater the surface area, more the adsorption

(3) Lower the temperature, more the adsorption

(4) Higher the pressure, more the adsorption **SC0089**

3. Among the following, the false statement is : **[JEE-MAIN-2019]**

(1) Latex is a colloidal solution of rubber particles which are positively charged

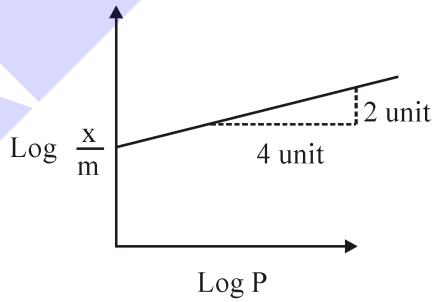
(2) Tyndall effect can be used to distinguish between a colloidal solution and a true solution.

(3) It is possible to cause artificial rain by throwing electrified sand carrying charge opposite to the one on clouds from an aeroplane.

(4) Lyophilic sol can be coagulated by adding an electrolyte. **SC0090**

4. Adsorption of a gas follows Freundlich adsorption isotherm. In the given plot, x is the mass of the gas adsorbed on mass m of the adsorbent at pressure p. $\frac{x}{m}$ is proportional to

[JEE-MAIN-2019]



(1) $P^{1/4}$ (2) P^2 (3) P (4) $P^{1/2}$ **SC0092**

5. Among the following, the INCORRECT statement about colloids is :

(1) They can scatter light **[JEE-MAIN-2019]**

(2) They are larger than small molecules and have high molar mass

(3) The range of diameters of colloidal particles is between 1 and 1000 nm

(4) The osmotic pressure of a colloidal solution is of higher order than the true solution at the same concentration **SC0096**

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6. A gas undergoes physical adsorption on a surface and follows the given Freundlich adsorption isotherm equation

$$\frac{x}{m} = kp^{0.5}$$

Adsorption of the gas increases with : **[JEE-MAIN-2019]**

(1) Decrease in p and decrease in T (2) Increase in p and increase in T

(3) Increase in p and decrease in T (4) Decrease in p and increase in T **SC0097**

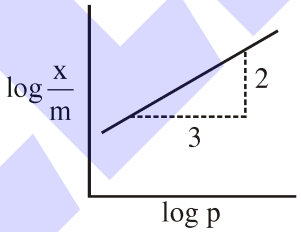
7. 10 mL of 1mM surfactant solution forms a monolayer covering 0.24 cm^2 on a polar substrate. If the polar head is approximated as cube, what is its edge length? **[JEE-MAIN-2019]**

(1) 2.0 pm (2) 2.0 nm (3) 1.0 pm (4) 0.1 nm **SC0099**

8. 0.27 g of a long chain fatty acid was dissolved in 100 cm^3 of hexane. 10 mL of this solution was added dropwise to the surface of water in a round watch glass. Hexane evaporates and a monolayer is formed. The distance from edge to centre of the watch glass is 10 cm. What is the height of the monolayer? [Density of fatty acid = 0.9 g cm^{-3}, $\pi = 3$] **[JEE-MAIN-2019]**

(1) 10^{-8} m (2) 10^{-6} m (3) 10^{-4} m (4) 10^{-2} m **SC0100**

9. Adsorption of a gas follows Freundlich adsorption isotherm x is the mass of the gas adsorbed on mass m of the adsorbent. The plot of $\log \frac{x}{m}$ versus log p is shown in the given graph. $\frac{x}{m}$ is proportional to : **[JEE-MAIN-2019]**



(1) $p^{3/2}$ (2) p^3 (3) $p^{2/3}$ (4) p^2 **SC0101**

10. The correct option among the following is : **[JEE-MAIN-2019]**

(1) Colloidal particles in lyophobic sols can be precipiated by electrophoresis.

(2) Brownian motion in colloidal solution is faster the viscosity of the solution is very high.

(3) Colloidal medicines are more effective because they have small surface area.

(4) Addition of alum to water makes it unfit for drinking. **SC0102**

11. The flocculation value of HCl for arsenic sulphide sol. is 30 m mole L^{-1}. If H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams, of H_2SO_4 in 250 ml required for the above purpose is _____.

(molecular mass of H_2SO_4 = 98 g/mol) **[JEE-MAIN-2020]**

SC0103

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12. As per Hardy-Schulze formulation, the flocculation values of the following for ferric hydroxide sol are in the order : **[JEE-MAIN-2020]**

(1) $AlCl_3 > K_3[Fe(CN)_6] > K_2CrO_4 > KBr=KNO_3$

(2) $K_3[Fe(CN)_6] < K_2CrO_4 < AlCl_3 < KBr < KNO_3$

(3) $K_3[Fe(CN)_6] > AlCl_3 > K_2CrO_4 > KBr > KNO_3$

(4) $K_3[Fe(CN)_6] < K_2CrO_4 < KBr=KNO_3=AlCl_3$ **SC0104**

13. For the following Assertion and Reason, the correct option is **[JEE-MAIN-2020]**

Assertion : For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7-9 elements.

Reason : The reactants are most strongly adsorbed on group 7-9 elements.

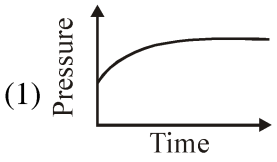
(1) Both assertion and reason are true but the reason is not the correct explanation for the assertion.

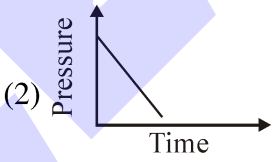
(2) Both assertion and reason are false.

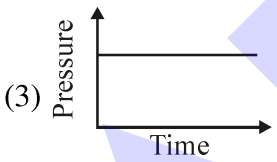
(3) Both assertion and reason are true and the reason is the correct explanation for the assertion.

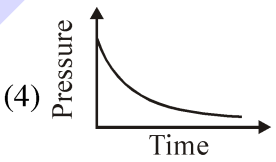
(4) The assertion is true, but the reason is false. **SC0105**

14. A mixture of gases O_2, H_2 and CO are taken in a closed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is : **[JEE-MAIN-2020]**

(1) 

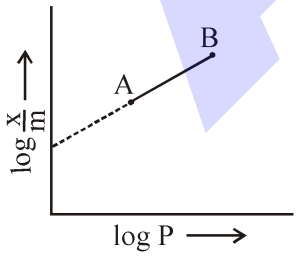
(2) 

(3) 

(4) 

SC0106

15. In Freundlich adsorption isotherm, slope of AB line is : **[JEE-MAIN-2021]**



(1) log n with (n > 1)

(2) n with (n, 0.1 to 0.5)

(3) $\log\frac{1}{n}$ with $(n<1)$

(4) $\frac{1}{n}$ with $\left(\frac{1}{n}=0 \text{ to } 1\right)$ **SC0119**

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16. The nature of charge on resulting colloidal particles when $FeCl_3$ is added to excess of hot water is

(1) Positive **[JEE-MAIN-2021]**

(2) Sometimes positive and sometimes negative

(3) Neutral

(4) Negative

SC0120

17. In Freundlich adsorption isotherm at moderate pressure, the extent of adsorption $\left(\frac{x}{m}\right)$ is directly proportional to P^x. The value of x is **[JEE-MAIN-2021]**

(1) zero (2) $\frac{1}{n}$ (3) 1 (4) ∞

SC0121

18. Which one of the following statements is FALSE for hydrophilic sols ? **[JEE-MAIN-2021]**

(1) Their viscosity is of the order of that of H_2O.

(2) The sols cannot be easily coagulated.

(3) They do not require electrolytes for stability.

(4) These sols are reversible in nature.

SC0122

19. Most suitable salt which can be used for efficient clotting of blood will be :-**[JEE-MAIN-2021]**

(1) $NaHCO_3$ (2) $FeSO_4$ (3) $Mg(HCO_3)_2$ (4) $FeCl_3$

SC0123

20. The charges on the colloidal CdS sol and TiO_2 sol are, respectively : **[JEE-MAIN-2021]**

(1) positive and positive (2) positive and negative

(3) negative and negative (4) negative and positive

SC0124

21. For the coagulation of a negative sol, the species below, that has the highest flocculating power is

(1) SO_4^{2-} (2) Ba^{2+} (3) Na^+ (4) PO_4^{3-} **[JEE-MAIN-2021]**

SC0125

22. The INCORRECT statements below regarding colloidal solutions is : **[JEE-MAIN-2021]**

(1) A colloidal solution shows colligative properties.

(2) An ordinary filter paper can stop the flow of colloidal particles.

(3) The flocculating power of Al^{3+} is more than that of Na^+.

(4) A colloidal solution shows Brownian motion of colloidal particles.

SC0126

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23. CO_2 gas adsorbs on charcoal following Freundlich adsorption isotherm. For a given amount of charcoal, the mass of CO_2 adsorbed becomes 64 times when the pressure of CO_2 is doubled.

The value of n in the Freundlich isotherm equation is _____ $\times 10^{-2}$. **[JEE-MAIN-2021]**

(Round off to the Nearest Integer) **SC0127**

24. Match List I with List II :

List-I **Example of colloids**	**List-II** **Classification**
(a) Cheese	(i) dispersion of liquid in liquid
(b) Pumice stone	(ii) dispersion of liquid in gas
(c) Hair cream	(iii) dispersion of gas in solid
(d) Cloud	(iv) dispersion of liquid in solid

[JEE-MAIN-2021]

Choose the most appropriate answer from the options given below **SC0128**

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

25. The conditions given below are in the context of observing Tyndall effect in colloidal solutions:

(A) The diameter of the colloidal particles is comparable to the wavelength of light used.

(B) The diameter of the colloidal particles is much smaller than the wavelength of light used.

(C) The diameter of the colloidal particles is much larger than the wavelength of light used.

(D) The refractive indices of the dispersed phase and the dispersion medium are comparable.

(E) The dispersed phase has a very different refractive index from the dispersion medium.

Choose the most appropriate conditions from the options given below: **SC0129**

(1) (A) and (E) only (2) (C) and (D) only **[JEE-MAIN-2021]**

(3) (A) and (D) only (4) (B) and (E) only

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EXERCISE # J-ADVANCED

1. Among the following, the surfactant that will from micelles in aqueous solution at the lowest molar concentration at ambient conditions is **[JEE 2008]**

(A) $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$ (B) $CH_3(CH_2)_{11}OSO_3^-Na^+$

(C) $CH_3(CH_2)_6COO^-Na^+$ (D) $CH_3(CH_2)_{11}N^+(CH_3)_3Br^-$ **SC0110**

2. Among the electrolytes Na_2SO_4, $CaCl_2$, $Al_2(SO_4)_3$ and NH_4Cl, the most effective coagulation agent for Sb_2S_3 sol is **[JEE 2009]**

(A) Na_2SO_4 (B) $CaCl_2$ (C) $Al_2(SO_4)_3$ (D) NH_4Cl **SC0111**

3. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) -

(A) Adsorption is always exothermic **[JEE 2011]**

(B) Physisorption may transform into chemisorption at high temperature

(C) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature.

(D) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation. **SC0112**

4. Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle. **[JEE 2012]**

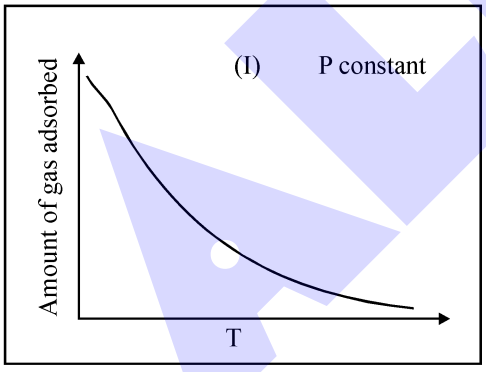
(A) Preferential adsorption of ions on their surface from the solution

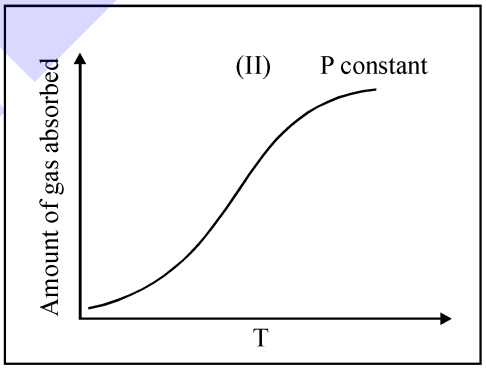
(B) Preferential adsorption of solvent on their surface from the solution

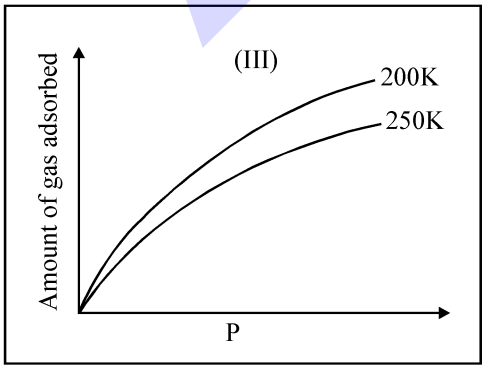
(C) Attraction between different particles having opposite charges on their surface

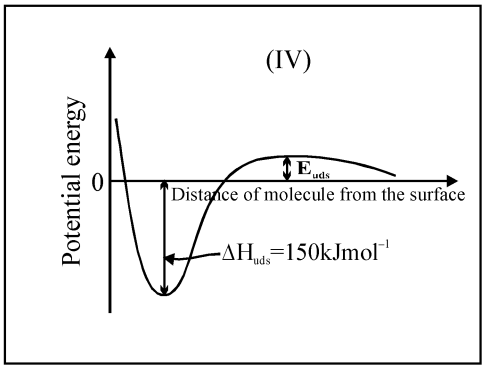
(D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles **SC0113**

5. The given graphs / data **I, II, III and IV** represent general trends observed for different physisorption and chemisorption processes under mild conditions of temperature and pressure. Which of the following choice(s) about **I, II, III and IV** is (are) correct ? **[JEE 2012]**









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(A) **I** is physisorption and **II** is chemisorption

(B) **I** is physisorption and **III** is chemisorption

(C) **IV** is chemisorption and **II** is chemisorption

(D) **IV** is chemisorption and **III** is chemisorption **SC0114**

6. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25ºC. For this process, the correct statement is - **[JEE 2013]**

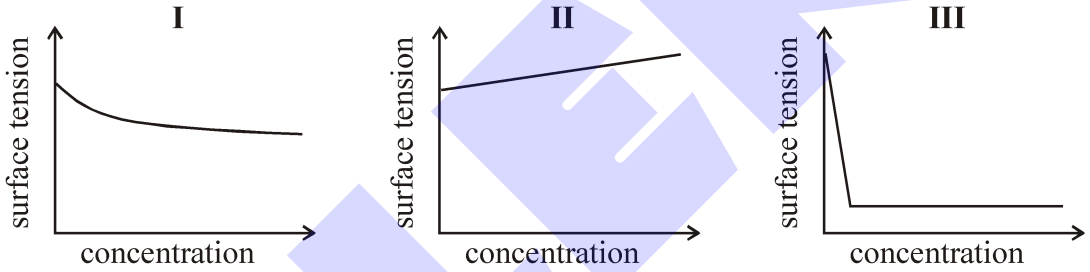
(A) The adsorption requires activation at 25ºC

(B) The adsorption is accompanied by a decrease in enthalpy

(C) The adsorption increases with increase of temperature

(D) The adsorption is irreversible **SC0115**

7. The qualitative sketches I , II and III given below show the variation of surface tension with molar concentration of three different aqueous solutions of KCl, CH_3OH and $CH_3(CH_2)_{11}OSO_3^-Na^+$ at room temperature. The correct assignment of the sketches is - **[JEE 2016]**



(A) **I :** KCl **II :** CH_3OH **III :** $CH_3(CH_2)_{11}OSO_3^-Na^+$

(B) **I :** $CH_3(CH_2)_{11}OSO_3^-Na^+$ **II :** CH_3OH **III :** KCl

(C) **I :** KCl **II :** $CH_3(CH_2)_{11}OSO_3^-Na^+$ **III :** CH_3OH

(D) **I :** CH_3OH **II :** KCl **III :** $CH_3(CH_2)_{11}OSO_3^-Na^+$

SC0116

8. The correct statement(s) about surface properties is (are) **[JEE 2017]**

(A) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium

(B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.

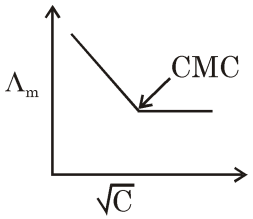
(C) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.

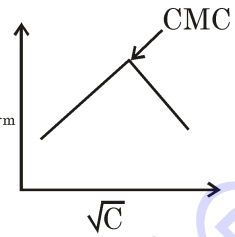
(D) The critical temperatures of ethane and nitrogen and 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature. **SC0117**

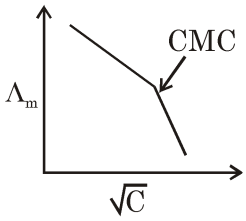
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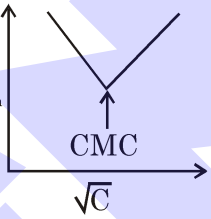
9. Molar conductivity (Λ_m) of aqueous solution of sodium stearate, which behaves as a strong electrolyte, is recorded at varying concentration(c) of sodium stearate. Which one of the following plots provides the correct representation of micelle formation in the solution ?

(Critical micelle concentration (CMC) is marked with an arrow in the figures.) **[JEE 2019]**

(1) 

(2) 

(3) 

(4) 

SC0118

10. The correct statement(s) related to colloids is(are) **[JEE 2021]**

(A) The process of precipitating colloidal sol by an electrolyte is called peptization.

(B) Colloidal solution freezes at higher temperature than the true solution at the same concentration.

(C) Surfactants form micelle above critical micelle concentration (CMC). CMC depends on temperature

(D) Micelles are macromolecular colloids.

SC0130

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ANSWER-KEY

EXERCISE # S-I

1. (200) **2.** (0.6) **3.** (1)

4. (5) **5.** (3) **6.** (30)

7. (5) **8.** (9) **9.** (600)

10 (4)

EXERCISE # S-II

1. (3) **2.** (1) **3.** (3)

4. (2) **5.** (2.4×10^6 cm^2/gm)

6. (4 sec) **7.** (6) **8.** $\frac{3}{140}$

9. (2.35×10^9) **10.** (10^4 times)

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	A	B	B	B	B	C	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	C	B	C	A	B	A	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	B	D	C	D	D	B	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	B	C	A	B	D	B	B	A	C

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,B,D	A,B	A,B,D	B,C,D	A,B	B,C,D	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A	D	B,D	A	A	A	C	B	C

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EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	4	4	3	1	2	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	0.36 to 0.38	4	4	4	4	1	2	1	4	4
Que.	21	22	23	24	25					
Ans.	2	2	17	4	1					

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A,B,D	A,D	A,C	B	D	B,D	3	B,C

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